Package ‘PhyloTraitDynamics’

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Title Companion Code for Phylogenetic Trait Dynamics

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Description Companion code for the article ``Phylogenetic dynamics of MRCA ages and empirical moments of a Brownian trait''. Public functions assume inputs consistent with the article's assumptions.

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`birth_death_brownian_simulate` *Simulate Birth-Death Brownian Empirical Moments*

Description

Simulates birth-death Brownian paths and records, on the requested time grid, the number of living lineages, the empirical mean of their traits, and their empirical variance.

Usage

```
birth_death_brownian_simulate(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  B,
  x0 = 0,
  seed = NULL,
  n_envelope = 4000,
  safety_factor = 1.05
)
```

Arguments

`birth`	Non-negative birth rate, either a numeric constant or a function of time.
`death`	Non-negative death rate, either a numeric constant or a function of time.
`sigma2`	Brownian variance parameter.
`time_end, time_step`	Time grid definition.
`B`	Number of simulated paths.
`x0`	Initial trait value.
`seed`	Optional random seed.
`n_envelope`	Number of points used by the thinning envelope.
`safety_factor`	Multiplicative safety factor for the thinning envelope.

Value

An object of class `"birth_death_brownian_simulation"` containing the time grid, matrices of empirical means, empirical variances and lineage counts, and a simulation summary.

`empirical_mean_compute_variance`
Variance of the Empirical Mean Under Birth-Death Brownian Dynamics

---

Description

Computes the theoretical variance of the empirical mean of Brownian traits, conditional on survival at each time.

Usage

```
empirical_mean_compute_variance(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  n_steps = 1200,
  tol_sing = 1e-07,
  li2_tol = 1e-12
)
```

Arguments

`birth`	Non-negative birth rate, either a numeric constant or a function of time.
`death`	Non-negative death rate, either a numeric constant or a function of time.
`sigma2`	Brownian variance parameter.
`time_end`, `time_step`	Time grid definition.
`n_steps`	Number of integration steps for numerical theory.
`tol_sing`, `li2_tol`	Numerical tolerances.

Value

A data frame with time, survival probability, empirical mean variance, and conditioning columns. The original script result is attached as attribute `"original_result"`.

`empirical_mean_plot_variance`
Plot Variance of the Empirical Mean

Description

Plots Monte Carlo summaries and/or theoretical values for the variance of the empirical mean.

Usage

```
empirical_mean_plot_variance(
  simulation = NULL,
  theory = NULL,
  main = NULL,
  xlab = "time",
  ylab = "Variance of empirical mean"
)
```

Arguments

`simulation`	Optional "`empirical_mean_simulation`" object.
`theory`	Optional data frame returned by `empirical_mean_compute_variance()`.
`main, xlab, ylab`	Base graphics labels.

Value

Invisibly returns a list containing `simulation`, `theory`, and `summary`.

---

empirical_variance_compute_expectation

Expected Empirical Variance Under Birth-Death Brownian Dynamics

---

Description

Computes the theoretical expectation of the empirical trait variance on a time grid for a birth-death process with Brownian trait evolution.

Usage

```
empirical_variance_compute_expectation(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  conditioning = c("none", "survival"),
  method = c("auto", "numeric", "closed_constant"),
  n_steps = 2000,
  tol = 1e-08,
  constant_tol = 1e-10,
  li2_tol = 1e-12
)
```

Arguments

`birth`	Non-negative birth rate, either a numeric constant or a function of time.
`death`	Non-negative death rate, either a numeric constant or a function of time.
`sigma2`	Brownian variance parameter.

`time_end`, `time_step`
: Time grid definition.

`conditioning`
: Conditioning used for the expectation. `"none"` is unconditional and `"survival"` conditions on `N(t) > 0`.

`method`
: Theory evaluation method. `"auto"` uses the closed constant-rate formula when the birth and death rates are constant on the time grid, and otherwise uses the numerical method. `"closed_constant"` requires constant birth and death rates.

`n_steps`
: Number of integration steps for numerical theory.

`tol`, `constant_tol`, `li2_tol`
: Numerical tolerances passed to the original implementation.

Value

A data frame with time, survival probability, expected empirical variance, conditioning, and method columns. The original script result is attached as attribute `"original_result"`.

`empirical_variance_plot_expectation`

Plot Expected Empirical Variance

Description

Plots simulated empirical variance paths, Monte Carlo summaries, and/or the theoretical expectation on a common set of axes.

Usage

```
empirical_variance_plot_expectation(
  simulation = NULL,
  theory = NULL,
  show_paths = TRUE,
  conditioning = c("none", "survival"),
  max_paths = Inf,
  alpha = 0.05,
  main = NULL,
  xlab = "time",
  ylab = "Empirical variance"
)
```

Arguments

`simulation`
: Optional `"empirical_variance_simulation"` object.

`theory`
: Optional data frame returned by `empirical_variance_compute_expectation()`.

`show_paths`
: Logical; whether to draw individual simulated paths.

`conditioning`
: Which expectation to plot.

`max_paths`
: Maximum number of simulated paths to draw.

`alpha`
: Transparency used for simulated paths.

`main`, `xlab`, `ylab`
: Base graphics labels.

Value

Invisibly returns a list containing `simulation`, `theory`, and `summary`.

`fixed_tree_compute_theoretical_summary`
Theoretical Summary on a Fixed Tree

Description

Computes theoretical summaries of empirical mean variance and empirical variance for Brownian traits on a fixed phylogenetic tree.

Usage

```
fixed_tree_compute_theoretical_summary(
  tree,
  sigma2 = 1,
  time_end = NULL,
  time_step
)
```

Arguments

`tree`	A phylogenetic tree of class "phylo".
`sigma2`	Brownian variance parameter.
`time_end`	End of the time grid. If `NULL`, the maximum tree time is used.
`time_step`	Time step used to define the time grid.

Value

A data frame with time, lineage counts, empirical mean variance, empirical variance expectation, empirical variance quartiles, and empirical variance variance. The tree, `sigma2`, and original result are attached as attributes.

`fixed_tree_plot_brownian_realization`
Plot a Fixed-Tree Brownian Realization

Description

Plots the simulated Brownian realization together with the fixed tree using the original plotting routine.

Usage

```
fixed_tree_plot_brownian_realization(
  x,
  n_bands = 10,
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05),
  cex = 1
)
```

Arguments

`x`	A `fixed_tree_brownian_realization` object.
`n_bands`	Number of time bands shown by the original plotting routine.
`time_band_col`	Color used for time bands.
`cex`	Character expansion factor passed to base graphics parameters.

Value

Invisibly returns `x`.

`fixed_tree_plot_theoretical_summary`

Plot a Fixed-Tree Theoretical Summary

Description

Plots fixed-tree theoretical summaries for empirical mean variance and empirical variance, optionally with the tree.

Usage

```
fixed_tree_plot_theoretical_summary(
  x,
  tree = NULL,
  show_tree = TRUE,
  main = NULL,
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05),
  band_col = grDevices::adjustcolor("red", alpha.f = 0.18),
  expectation_col = "blue",
  median_col = "red",
  mean_variance_col = "darkgreen",
  cex = 1
)
```

Arguments

`x`	A `fixed_tree_theoretical_summary` object, as returned by `fixed_tree_compute_theoretical_`
`tree`	Optional phylogenetic tree of class `"phylo"`. If `NULL`, the tree attached to `x` is used when available.
`show_tree`	Logical; whether to show the tree panel.
`main`	Optional main title.
`time_band_col`	Color used for time bands.
`band_col`	Color used for the empirical-variance interquartile band.
`expectation_col`	Color used for the empirical-variance expectation.
`median_col`	Color used for the empirical-variance median and quartiles.
`mean_variance_col`	Color used for the empirical-mean variance.
`cex`	Character expansion factor passed to base graphics parameters.

Value

Invisibly returns x.

---

fixed_tree_simulate_brownian_realization

Simulate Brownian Traits on a Fixed Tree

Description

Simulates one Brownian trait realization on a fixed phylogenetic tree and records empirical mean and empirical variance through time.

Usage

```
fixed_tree_simulate_brownian_realization(
  tree,
  sigma2 = 1,
  time_step,
  seed = NULL
)
```

Arguments

tree	A phylogenetic tree of class "phylo".
sigma2	Brownian variance parameter.
time_step	Time step used for the simulation output.
seed	Optional random seed.

Value

An object of class "fixed_tree_brownian_realization" containing the input tree, simulation data frame, time series, parameters, and seed.

---

mrca_age_compute_distribution

MRCA Age Distribution at One Time

Description

Computes the distribution and summary statistics of the most recent common ancestor age at a fixed time under a birth-death process.

Usage

```
mrca_age_compute_distribution(
  birth,
  death,
  time,
  time_step = NULL,
  probs = c(0.25, 0.5, 0.75),
  n_internal = NULL
)
```

Arguments

`birth`	Non-negative birth rate, either a numeric constant or a function of time.
`death`	Non-negative death rate, either a numeric constant or a function of time.
`time`	Observation time.
`time_step`	Internal time step. If `NULL`, a default step is derived from `time`.
`probs`	Quantile probabilities to report.
`n_internal`	Optional number of internal subintervals; when supplied it overrides `time_step`.

Value

An object of class `"mrca_age_distribution"` with distribution grid, summary statistics, requested probabilities, and the original result.

`mrca_age_compute_dynamics` — *MRCA Age Distribution Dynamics*

Description

Computes MRCA age distribution summaries over a time grid under a birth-death process.

Usage

```
mrca_age_compute_dynamics(
  birth,
  death,
  time_end,
  time_step,
  probs = c(0.25, 0.5, 0.75),
  n_internal_per_step = 10L,
  include_zero = TRUE
)
```

Arguments

`birth`	Non-negative birth rate, either a numeric constant or a function of time.
`death`	Non-negative death rate, either a numeric constant or a function of time.
`time_end, time_step`	Time grid definition.
`probs`	Quantile probabilities to report.
`n_internal_per_step`	Number of internal integration substeps per output time step.
`include_zero`	Logical; whether to include time zero in the original dynamics computation.

Value

An object of class `"mrca_age_dynamic"` containing the summary data frame, distribution grids, probabilities, and original call.

`mrca_age_plot_dynamics` *Plot MRCA Age Distribution Dynamics*

Description

Plots MRCA age summaries over time, including median, optional interquartile band, and optional expectation.

Usage

```
mrca_age_plot_dynamics(
  x,
  show_iqr = TRUE,
  show_expectation = TRUE,
  main = NA,
  xlab = "Time",
  ylab = "MRCA age",
  add_legend = TRUE,
  ...
)
```

Arguments

`x`	An `"mrca_age_dynamic"` object.
`show_iqr`	Logical; whether to draw the interquartile band.
`show_expectation`	Logical; whether to draw the expectation curve.
`main, xlab, ylab`	Base graphics labels.
`add_legend`	Logical; whether to draw the plot legend.
`...`	Additional arguments passed to `graphics::plot()`.

Value

Invisibly returns `x`.

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